

VIATRIS INC. Analysis + Mitigation



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Attack Surface

Viatrix Inc. has an expansive attack surface of pharmaceutical research labs, manufacturing sites, and business locations around the globe. This includes:

- Pharmaceutical R&D and manufacturing systems (ICS) and equipment that run application code
- IoT sensors within R&D and manufacturing facilities
- Company employee IoT devices
- Internal and external data centers and servers
 - Cloud, such as AWS or Azure
 - Physical databases/centers in-house or at a different location
- Website and website 3rd and 4th party widgets and code
 - OneTrust
 - Optanon
 - FullCalendar
 - Disqus
 - ASP.NET
 - Vimeo
 - jQuery
 - Backbone.js
 - Typeahead.js
 - MSN/Bing Webmaster
 - Workday CMS
- Ports in use on the network
- Network IDS/IPS/EDS servers and the computers they run on
- Login interfaces for SaaS and IaaS products installed on systems that include but are not limited to:
 - SAP
 - AWS
 - FLEX
 - Salesforce
 - Azure - Data Factory, Synapse
 - Rapid Response
 - MS Office Suite
- Applications code including the above SaaS products installed on devices that include but are not limited to:
 - LabWare
 - PK Statistical Program
 - Sigma Plot
 - Lotus Notes
 - AutoCAD
 - Empower
 - Trackwise
 - Power BI

- Salesforce
 - Veeva
 - Documentum
 - Power Apps, Automate
 - Tabular Editor
- APIs
- Local storage devices
- Files

The tools and applications listed above are deployed globally, so all IoT, employee devices, and company hardware are inclusive of this attack surface.

According to many of their job listings found in one of the prior sections of this report, much of the software they use within their laboratories and business operations are often legacy software that has since been purchased and renamed by another company. This is particularly dangerous for Viatrix because these systems have major CVEs and are likely unpatched and, in turn, increases the depth and extent of damage a potential threat actor can impose on the organization.

Threats and Threat Actors

Crime Syndicate Hackers

Crime syndicate-associated hackers will be one of the most dangerous of threat actors due to their technical prowess and intent to do harm. This is especially true for syndicate members who are driven to steal R&D information to sell to competitors on the black market. This could be especially disrupting for Viatris' operations should their proprietary data be leaked and give competitors an unfair advantage. These attackers could also potentially be insider threats masquerading as legitimate employees, should their records be clean enough.

Insider Threats

Insider threats are likely the most dangerous of all other threat actors. Many will have access to legacy systems that are running old hardware that has unpatched vulnerabilities. Given the extensive list of applications that Viatris lists on its job applications that are outdated or have since been purchased and renamed, this will be the primary method of causing harm. The question only becomes how well an individual can get away with it or if they only want revenge and do not care about the repercussions.

Cartels

Cartels are a very interesting threat actor type but may prove to be difficult to deal with given the money and influence they can have on local law enforcement in a given area. They may be able to hire very adept hackers and use physical means of influence (i.e., kidnapping and subsequently physical threats) to gain information. Cartels may not be the most likely to attack a pharmaceutical company, but they would have a great influence should they decide to target specific Viatris employees either for recruitment or extortion purposes.

Hacktivists

Hacktivists may target specific individuals inside the company to cause harm. In particular, the CIO of Viatris had been accused of insider trading and is still involved with the company. Additionally, a former board member was charged with manipulating tax documents and will be sentenced later this October. Should a hacktivist, especially one that may have personal investments in the company, feel they are righteous for attacking due to these reasons, cause Viatris to suffer. This could occur for any number of reasons, such as individuals who do not believe in western medicine or treatments.

Untargeted Attackers

Untargeted attackers are the least likely of this group due to the extensive work they would need to do to make widespread impact that also impacted Viatris. Untargeted attackers are likely individuals trying to attack 3rd and 4th party web infrastructure for website defacement for fun or to cause distress. This could also encompass individuals who want to impact internet infrastructure as a whole but are unlikely as these individuals do not have motives as strong or technical skills as extensive as the others above in this list. They also likely do not have as much time.

Intelligence Requirements

The following Intelligence Requirements have been chosen based on the needs and the threats identified specifically to our organization.

1. Intelligence Requirement #1: Monitor current events regarding attacks on industrial control systems including current events, threat actors, and publicly disclosed system vulnerabilities.
 - a. Priority Intelligence Requirement #1-1: Identify recent and historical attack patterns on Industrial Control Systems
 - b. Priority Intelligence Requirement #1-2: Identify information avenues for being alerted to and gathering Industrial Control System equipment software and firmware updates
2. Intelligence Requirement #2: Stay up-to-date on current phishing and social engineering attack trends
 - a. Priority Intelligence Requirement #2-1: Evaluate the likelihood of a successful phishing campaign being launched against the company over the next business quarter
 - b. Priority Intelligence Requirement #2-2: Gather intel on techniques being used by malicious actors to bypass two-factor-authentication
3. Intelligence Requirement#3: Monitor for company domain information, or employee credential information, available online
 - a. Priority Intelligence Requirement #3-1: Review public sources for email compromises associated with company domains
 - b. Priority Intelligence Requirement #3-2: Review dark-web sources for any credentials or other intelligence about the company being sold

Data Sources

The following Data Sources are recommended for acquiring Cyber Intelligence to support our Intelligence Requirements.

Internal Data Sources:

Internal Data Source 1: Logging/analysis of internal networks (ICS network, business network, DMZ, VPN)	
Requires Funding?	Yes: Using AlienVault IDS sensors licensing
Frequency of Source Feeds?	Analysis in real-time, logs backed up daily
Format of Source Data?	OTX (AlienVault Proprietary), supports CSV
Supports Information Sharing?	Yes: True Positives are fed into the AlienVault database
Initial Admiralty Code?	B2: Usually Reliable, Probably True

Internal Data Source 2: “Report Phishing” option feature feed on company email	
Requires Funding?	No
Frequency of Source Feeds?	Hourly
Format of Source Data?	Email
Supports Information Sharing?	N/A
Initial Admiralty Code?	C3: Fairly Reliable, Possibly True

External Data Sources

External Data Source 2: Open-Source Threat Intelligence Feed(s): VirusTotal, MITRE, CISA	
Requires Funding?	No
Frequency of Source Feeds?	Daily
Format of Source Data?	Reports, signatures
Supports Information Sharing?	Yes
Initial Admiralty Code?	B1: Usually reliable, Confirmed by Other Sources

External Data Source 2: Dark Web Data Source (DarkOwl)	
Requires Funding?	Yes
Frequency of Source Feeds?	Daily
Format of Source Data?	JSON
Supports Information Sharing?	No
Initial Admiralty Code?	B2: Usually Reliable, Probably True

Threat Analysis Workflows

1. STRIDE - Developed by microsoft, it is a acronym of certain violation of the CIA triad:
 - a. Spoofing of an account or computer
 - b. Tampering of data
 - c. repudiation which stops you from accessing links
 - d. Information disclosure which violates
 - e. Denial of service to users
 - f. Elevation of privilege on accounts
2. PASTA - stands for Process for attack Simulation and Threat Analysis
 - a. Define objectives
 - b. Define Technical Scope
 - c. Application Decomposition
 - d. Threat Analysis
 - e. Vulnerability & Weakness analysis
 - f. Attack modeling
 - g. Risk & impact analysis
3. Persona Non Grata
 - a. Create profiles with consideration for attackers and their:
 - i. Motivations
 - ii. Abilities
 - b. Brainstorm about targets' likely attack mechanisms
4. DREAD
 - a. Assess risks in the following 5 dimensions:
 - i. Damage potential
 - ii. Reproducibility
 - iii. Exploitability
 - iv. Affected users
 - v. Discoverability
5. Trike - an open source for threat model and risk management
 - a. Identify assets, define requirements
 - b. Create diagrams
 - c. Identify risks and threats
 - d. Mitigate threats
 - e. Validate the model
6. General workflow (SEI, 73)
 - a. Know your environment
 - b. Identify and understand gaps and intelligence requirements (IRs, PIRs, SIRs)
 - c. Verify internal and external data sources
 - d. Conduct tactical analysis to answer what/ where/when/how questions regarding threats, attacks, incidents, vulnerabilities, or other unusual network activity for implementing controls

- e. Conduct operational analysis, adding context (threat actors, campaigns) to existing tactical intelligence; starting to answer the who and why behind threats
- f. Enhance mid- to senior-level leadership decisions regarding non-immediate but near-term (weekly-quarterly) business process and operational decisions.
- g. Leadership provides feedback on analysis and recommendations

Strategic Analysis Workflows

General Workflow - must be clear and repeatable with timeline, roles, and responsibilities

1. Know your environment (aka how systems interact and talk to each other)
2. Identify and understand cyber risk gaps and intelligence requirements
3. Fuse Threat Analysis with other external and non-traditional data sources like news sources or social media
4. Depending on data collected, work with a data science team to identify larger trends or anomalies in data collected and collaborate with ISACs to understand a large issue.
5. Perform structured analytical techniques and human-centered design activities as needed (ex: tabletop exercises, collaborative teaching)
6. Provide analytical assessments (e.g risk assessment) based on threat actor potential, organizational exposure, and organizational impact of threat
7. Analyze current and future technologies and geopolitics that may positively/negatively impact the organization and industry
8. Enhance executive leader decision making by answering IRs and providing intelligence pertaining to organizational strategic risks regarding financial health, brand, stature, and reputation
9. Leadership provides feedback on information given.

(SEI, 2019 p.73)

Fusion Center Job Roles and Responsibilities

Fusion center - Mature

Security Operations

- Certified Vulnerability Assessor (CVA) - examines vulnerabilities hackers look for when trying to compromise a system
- Network Operations analyst - is incharge of planning, executing, and managing network services and systems, including hardware and virtual environments.
- Technical Support - responsible for providing technical support to customers or coworkers who require assistance with IT systems such as hardware and software or understanding organizational rules and processes such as not downloading freemium.
- Cyber Defense incident responder - responds to cyber incidents and provides guidance and documentation for the incident.

Security Engineering and Asset Security

Host and network security

- Malware and forensics analysts - Responsible for assessing malware during computer security incidents or research to provide guidance during an incident to remediate system and network vulnerabilities.
- Security Guard - a person hired by a government or private party to protect the employing party's assets from a variety of threats by enforcing proactive and preventative measures
- IT auditor - combs through IT assets to make sure the use is relevant to the organization.
- Identity access Manager - responsible for the onboarding and retiring of access to databases, servers and accounts
- Security engineering - in charge of preventing possible threats, weakness, and cyberattacks against an organization's data and information technology infrastructure.

Program Management

- Program management office
 - Mission Assessment specialist (Project Manager) - the main point of contact for a team and in charge of the team's particular projects.
- Governance, Risk, and Compliance
 - GRC Analyst - in charge of monitoring and analyzing information security policies, procedures, and controls within an organization to meet industry compliance.
 - Cyber Legal Advisor - provides legal advice and recommendations based on cybersecurity law
 - Privacy Officer/Compliance Manager - supports and manages privacy compliance program and privacy program staff, supporting privacy compliance, governance/policy, and incident response needs of privacy and security executives and their teams.

Internal and external relationships

Partner integration planner - Assists companies through mergers and acquisitions. Integration managers can assist both businesses involved in a merger or acquisition, providing advice, managing projects and facilitating meetings.

Business Development and Marketing

Business Analyst - compile and analyze company data to identify gaps and recommend controls. They work across their clients' teams to eliminate redundancies and improve efficiency. (Tompkins, 2023)

Cyber Intelligence

- Threat analysis
 - Threat/warning analysis - A professional dedicated to the identification, evaluation, and dissemination of potential risks that could impact an organization and/or community,
 - Digital Evidence Analysis - Responsible for identifying, collecting, examining, and preserving digital evidence using controlled and documented analytical and investigative techniques such as chain of command and hashing.
 - Cyber defense analyst - uses data collected from a variety of cyber defense tools within their environments (e.g. network traffic logs, firewalls, and IDS alert) to analyze events that occur for mitigating threats.

<https://www.cisa.gov/careers/work-roles/cyber-defense-analyst>

- Collection management
 - Cyber Intelligence Planner - develops comprehensive intelligence strategies to meet cyber operations requirements. Collaborates with cyber operations planners to determine, verify, and access data that needs to be gathered and analyzed. Chooses targets and carries out attacks while also organizing intel gathering (*Cyber Intelligence Planning [NICE Framework Work Role]*, n.d.).
 - cyber operations planners -developing cybersecurity operations plans; participating in targeting selection, validation, and synchronization; and enabling integration during the execution of cyber actions (*Cyber Operations Planning [NICE Framework Work Role]*, n.d.).
- Strategic Analysis
 - AR source Collection Manager - Responsible for indemnifying proper environments and authorities for intelligence requirements; Determines capabilities of available intelligence collection assets; constructs and disseminates intelligence collection plans; and oversees the execution of intelligence collection tasks to ensure effective execution plans (*Cyber Operations Planning [NICE Framework Work Role]*, n.d.).
 - Strategic Analyst - provide information for better decisions, address challenges and identify new ways for the company to grow (*Strategy Analyst Overview*, n.d.).
 - Geopolitical Analysts investigate, examine, and evaluate political structures of various geographical regions. You collect and analyze data, evaluate the effects of laws and

policies, monitor current events, and forecast economic, social, and political trends (*What Is a Geopolitical Analyst and How to Become One*, n.d.).

- Intelligence Analyst - analyze information and allow us to evaluate, mitigate, and eliminate threats (*Intelligence Careers at the FBI Intelligence Analysts*, n.d.).

Physical Security

- Security Guard - in the event of an emergency this person protects an organization or asset. Their duties are to monitor a specific area, evict trespassers and write reports detailing any occurrences during their shift. (*Security Guard Job Description: Top Duties and Qualifications*, n.d.)
- Physical security - responsible for physical security measures to protect property, people, and assets to reduce the risk of crimes. (Albert, 2023)

Technology Development & Integration

- Data science and machine learning
- Data Analyst - gathers, cleans, and studies data sets to help solve problems (Staff, 2024).
- Machine Learning Engineer - assess, produce, and optimize AI solutions by using capabilities and knowledge of conventional ML approaches (*Professional ML Engineer Certification | Learn | Google Cloud*, n.d.).

Software Application and Development

- Knowledge Management - is a person with the responsibility of identifying, organizing, storing and disseminating information within an organization to proper workflows and intelligence.

Tools

The following tools are recommended for a strong cybersecurity posture and support of the chosen Intelligence Requirements.

Network Monitoring:

Name	AlienVault IDS
Services Provided	● Real-Time Network Monitoring ○ Deep-Packet Inspection ● Data Logging and Backups ● Pairs with OTX (Open Threat Exchange) Intelligence Feed
Cost	Annual license fee: higher package options provide more sensors for deployment

Name	WatchGuard Firewall Hardware/Software
Services Provided	● Network Traffic Controls ● VPN Access Controls ● Internet traffic reports
Cost	Annual license fee per Firewall Device

Intelligence Feeds:

Name	AlienVault OTX
Services Provided	● Live Data feed for updated attack signatures and IoCs ● Pairs with OTX IDS
Cost	Annual license fee

Name	DarkOwl
Services Provided	● Online monitoring of Dark-Web resources for intelligence related to the company
Cost	Annual license fee

Anti-virus / Anti-malware:

Name	Bitdefender Antivirus
Services Provided	• Antivirus • Daily updated signatures
Cost	Annual license fee dependent on number of PCs

Name	Tripwire
Services Provided	• Anti-ransomware alerts
Cost	Open-Source version available; recommend commercial version

Data-loss Prevention:

Name	Forcepoint Data Loss Prevention
Services Provided	• Data-Loss Prevention Monitoring
Cost	Annual license fee

Monitoring Activities & Indicators to Track

1. Monitoring ICS and SCADA Systems Traffic

Implementing real-time ICS traffic monitoring, flagging any suspicious activities/traffic patterns. Setup alerts for unauthorized changes in configurations (ex: change in account privileges). Alerts are also triggered when unauthorized software or hardware are installed/inserted into/onto systems via SCADA.

- Unusual activity spikes in ICS network traffic
- Repeated or failed login attempts to SCADA systems
- Communication between ICS components and external IP addresses
 - This should be **internal-only**

2. Monitoring Signs of Data Exfiltration

Using a data loss prevention system (DLP) to monitor any signs of data exfiltration with flagging of any sensitive/critical files out of the network, cloud, or on-prem data centers.

- Employees accessing critical R&D data when they do not require access, particularly when done from a different location or odd timings.
- Any attempts for copying or transferring sensitive, confidential research files.
- Many packets leaving the network from an untrusted IP

3. Insider Threat Monitoring

Performing user behavior analytics (UBA systems) to track any suspicious old or newly created users.

- Abnormal login/logout times
- Abnormal lateral movement or requests to access within or between critical systems

4. Detection of Suspicious Network Traffic

Detecting and analyzing any anomalous behaviors within the network, tracking high usage of GPUs, CPUs, RAM, or Memory which can suggest malware infections.

- Abnormal spikes in traffic, or outgoing data packets to unfamiliar IPs
- High frequency of failed login attempts in short duration of time

5. Monitoring Cloud Activity through Access Logs

Through cloud logging services, monitoring of all cloud activity for unusual access to applications and data. Paying attention to specific cloud services, (open S3 buckets, MS Sentinel flagged reports). Flagging abnormal data transfer activity between internal cloud storage and external locations.

- Sudden changes in access permissions
- Increased amount of data transfer from cloud infrastructure

Fusion Center Collaboration Partners

Fusion Center Collaboration Partners allows Viatris to divide and conquer its search for threats on the horizon by taking shared information from these sources below.

Partner	Type	Benefits
Health-ISAC	Information Sharing and Analysis Center (ISAC)	Shares intel on attacks specific to the healthcare industry.
Pharma-ISAC	ISAC	Shares intel on attacks specific to the pharmaceutical industry.
MITRE ATT&CK Framework	Framework	Commonly identified Intelligence on adversary TTP to map possible attack vectors.
Recorded Future	Threat Intelligence Vendor	Monitors dark web activity to provide insights on leaked credentials and cyber campaigns targeting pharma.
US-CERT	CERT	National-level intelligence and support for incident response and critical infrastructure protection.
Industry CERTs	CERT	Provides sector-specific intelligence and helps coordinate responses to incidents.
Carnegie Mellon CyLab	Research Institution	Offers insights into cutting-edge research on new vulnerabilities and threat mitigation.
University of Maryland Cyber Institute	Research Institution	Provides advanced research on supply chain risks and cyber risk modeling.
ENISA	Cybersecurity Alliance	Offers intel on threats targeting European pharmaceutical and healthcare companies.

PSCI	Pharmaceutical Supply Initiative	Collaborative organization to share information about responsible supply chain management for better economic and social outcomes
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Social Engineering & Insider Threats Plan

Mitigating Social Engineering Attacks

1. Intelligence gathering and analysis
 - a. From the SIR workflows, aggregate necessary information for analysis for the CTIC
 - b. Consult online repositories for phishing sites regularly and add known phishing sites to a list of sites that will be blocked on the firewall
 - c. Create reports that show trends or changes in efficacy of the phishing training and awareness program
2. Policies and procedures
 - a. Establish clear onboarding and offboarding procedures
 - i. Do not retain old logins or email addresses when an individual no longer works for the organization. Dispose of them immediately but retain email logs for 1 year.
 - b. Require employees to flag spam and phishing emails
 - c. Send a notification to SOC when phishing emails slip past the spam filtering for remediation
3. Technical controls
 - a. Implement access control based on severity of potential damage to asset
 - i. Ex: On-prem CTIC database access: require badge access and/or biometric access when necessitated
 - ii. Log all logins and logouts on IAM
 - iii. Require password protected VPNs through Viatrix' SSO for remote work
 - b. Grant least privileges
 - i. If the privilege is not needed to perform the job, do not grant it
 - ii. This is especially important for on-prem vendors that require access to the building
 - c. Audit IAM systems and analyze logs within the CTIC
 - d. Block known phishing sites on host and network firewalls
4. Employee training and awareness
 - a. Establish a regular training and awareness program
 - i. Provide phishing training when onboarding and recurring bi-annually
 - ii. Ensure training covers vishing, smishing, and whaling specifically for the CTIC
 - b. Occasionally send out phishing emails to test efficacy of training and adjust where necessary
 - c. Do not punish employees for failing to recognize spam/phishing emails
5. Incident response plan
 - a. If a phishing email is found, report the email and notify colleagues
 - b. If an employee clicks on an attachment, let the antivirus do its work, but retrieve the computer to reverse-engineer the malware
 - c. Analyze the malware and its purpose
 - d. Create a report and take it into consideration for future reports to send to the SOC and add a tag of sorts on that specific type of file entering the network

Mitigating Insider Threats

1. Intelligence gathering and analysis
 - a. Prior to hiring, conduct extensive background checks and call references on the individual's resume before hiring
 - b. Aggregate IAM logs and analyze for anomalous activity
 - c. Audit when necessary
 - d. Monitor files going in and out of the network and hosts and flag important files for extrusion attempt alerts
 - e. Implement host-based firewall rules to block malicious sites and an IDS
 - i. Exceptions can be made for malicious sites for the purpose of malware analysis when done on a secured VM that cannot leak onto the main device, but must be documented and approved beforehand
2. Policies and procedures
 - a. Do not retain old login credentials for employees no longer at the organization. No exceptions.
 - b. Keep old email logs for a minimum of 1 year
 - c. When offboarding, remove credentials as soon as intent to layoff/fire occurs unless stated otherwise
3. Technical controls
 - a. Grant least access to all individuals on network; do not give individuals access to information they do not need to use for their job
4. Employee training and awareness
 - a. In Monthly group meetings, different teams will share their insights on new threats and mitigation strategies
 - b. Make employees aware of an anonymous tip system if they suspect their coworker may be acting unethically
 - i. Share that there is a direct connection to the CISO if the CTIC manager is compromised
5. Incident response plan
 - a. If malicious activity is found, forward the information to the CTIC manager, HR manager, and begin the offboarding process once approved
 - b. If suspicious activity is found, notify the CTIC manager (or CISO if the manager is compromised) and begin an investigation
 - i. Pull all log data from user accounts and from employee hardware
 - ii. Aggregate the logs and correlate actions to determine if there is malicious activity or if it breaks CTIC policy

Reporting Structure

The reporting structure designates the types of report, who they will be given to, and when.

1. Daily Threat Activities

A daily threat report created to brief the Security Operations Center (SOC) on new/emerging threats found within the past day. Combined with critical news from ISACs about new threats & vulnerabilities.

For SOC analysts

2. Weekly Vulnerabilities and Attack Summary

A weekly summary of new/emerging threats to be shared with the CTIC's closest and most immediate collaborators in Viatrix including detected attacks against other pharmaceutical companies.

For CISO, SOC Manager and their respective team

3. Monthly Threat & Vulnerability Report

A monthly evaluation of new and emerging threats gathered from the CTIC to be shared with technical positions to adjust posture as needed. CTIC may also make recommendations for policy/procedure.

For CISO, SOC, Risk Management Team

4. Quarterly Threat & Vulnerabilities Report

Quarterly evaluation of new and old risks to share with the largest stakeholders to aid in future decision-making for its cybersecurity posture and risk appetite.

For C-Suite executives, SOC, Risk Management Team, Board Members, Shareholders

5. Incident Response Reports

Reports created during an active cyber incident detailing the scope of the threat, attack vector, scope of the breach, affected systems, and mitigation steps followed. Best practices for similar future events included in case attack patterns and network control instances match.

For Incident Response, CISO, Legal Team, Host and Network security team, Public Relations

References

- Albert. (2023, August 27). Read info about how to become Physical Security Analyst, complete with Physical Security Analyst job description, skills, competencies and also Physical Security Analyst interview questions. *gomediajobs.com*.
<https://gomediajobs.com/physical-security-analyst/>
- All-Source Collection Management [NICE Framework Work role]*. (n.d.). National Initiative for Cybersecurity Careers and Studies. <https://niccs.cisa.gov/workforce-development/nice-framework/work-role/all-source-collection-management>
- Cyber Intelligence Planning [NICE Framework Work role]*. (n.d.). National Initiative for Cybersecurity Careers and Studies. <https://niccs.cisa.gov/workforce-development/nice-framework/work-role/cyber-intelligence-planning>
- Cyber Operations Planning [NICE Framework Work role]*. (n.d.). National Initiative for Cybersecurity Careers and Studies. <https://niccs.cisa.gov/workforce-development/nice-framework/work-role/cyber-operations-planning>
- How to become an Integration Manager (Plus common skills)*. (n.d.). Indeed.
<https://www.indeed.com/career-advice/finding-a-job/how-to-become-integration-manager>
- Intelligence careers at the FBI intelligence analysts*. (n.d.). FBI jobs.
<https://fbijobs.gov/intelligence-analysts>
- Professional ML Engineer Certification | Learn | Google Cloud*. (n.d.). Google Cloud.
<https://cloud.google.com/learn/certification/machine-learning-engineer>
- Security Guard Job Description: Top duties and qualifications*. (n.d.). Indeed.
<https://www.indeed.com/hire/job-description/security-guard>

SEI. (2019, p. 73). *Cyber Intelligence Tradecraft Report: The State of Cyber Intelligence Practices in the United States (Study Report and Implementation Guides)*.

Insights.sei.cmu.edu. <https://insights.sei.cmu.edu/library/cyber-intelligence-tradecraft-report-the-state-of-cyber-intelligence-practices-in-the-united-states-study-report-and-implementation-guides/>

Staff, C. (2024, April 1). *What does a data analyst do? Your 2024 career guide*.

Coursera. <https://www.coursera.org/articles/what-does-a-data-analyst-do-a-career-guide>

Strategy Analyst Overview. (n.d.). Glassdoor.

https://www.glassdoor.com/Career/strategy-analyst-career_KO0,16.htm

Tompkins, J. (2023, May 8). *What does a business analyst do? Everything you should*

know. Forbes Advisor. <https://www.forbes.com/advisor/education/business-and-marketing/what-does-a-business-analyst-do/>

What is a geopolitical analyst and how to become one. (n.d.). Ziprecruiter.

<https://www.ziprecruiter.com/career/Geopolitical-Analyst/What-Is-How-to-Become>